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Batch : DA | ML – DAF50
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Data Analytics Mini Project

Data Link : https://www.kaggle.com/datasets/farshidbahrami021/road-accident-dataset/data

About Raw Dataset

The Road Accident Analysis dataset contains **1,610 accident records** collected from various states across the United States. Each record represents a unique road accident event and includes information related to accident location, date and time, weather conditions, road characteristics, vehicle details, driver demographics, and emergency response metrics.

Screen short for raw uncleaned data is given below.

ID	State	Date	Day_of_Week	Time_of_Day	Weather_Conditions	Road_Conditions	Light_Conditions	Type_of_Road	Type_of_Junction	Type_of_Accident	Vehicle_Type	Driver_Age_Group	Num_Vehicles_Involved	Num_Cas
165433	Connecticı	########	Tuesday	19:59:02	Rainy	Muddy	Daylight	Street	T-Junction	Run-off-road	Car	Adult	2	
165434	Massachus	########	Wednesday	21:28:51	Sunny	Muddy	Daylight	Highway	Intersection	Side collision	Pedestrian	Adult	1	
165435	Wyoming	########	Thursday	18:04:26	Snowy	Snowy	Dawn	Street	Underpass	Cyclist involved	Bicycle	Young Adult	3	
165436	Hawaii	########	Sunday	16:51:02	Snowy	Icy	Daylight	Street	Intersection	Run-off-road	Car	Senior	5	
165437	Oklahoma	########	Tuesday	14:19:19	Foggy	Icy	Night	Rural	T-Junction	Run-off-road	Car	Adult	2	
165438	Minnesota	########	Monday	00:14:21	Foggy	Snowy	Dawn	Urban	T-Junction	Head-on collision	Pedestrian	Teenager	4	
165439	Wyoming	########	Saturday	10:21:34	Stormy	Icy	Daylight	Street	Crossroads	Rollover	Pedestrian	Teenager	1	
165440	Minnesota	########	Tuesday	02:01:40	Rainy	Icy	Daylight	Urban	Underpass	Rear-end collision	Motorcycle	Senior	4	
165441	Nevada	########	Wednesday	06:09:08	Rainy	Wet	Dusk	Highway	Overpass	Rear-end collision	Car	Teenager	4	
165442	North Dak	########	Tuesday	22:33:51	Sunny	Icy	Dawn	Street	Overpass	Pedestrian involved	Pedestrian	Senior	1	
165443	North Dak	########	Tuesday	05:49:45	Rainy	Wet	Dawn	Highway	Roundabout	Rear-end collision	Car	Adult	1	
165444	Iowa	########	Saturday	12:48:53	Stormy	Muddy	Night	Rural	Underpass	Rollover	Truck	Adult	1	
165445	Rhode Isla	########	Tuesday	08:18:51	Sunny	Icy	Daylight	Freeway	T-Junction	Rollover	Pedestrian	Adult	2	
165446	Delaware	########	Monday	21:51:12	Sunny	Icy	Dawn	Highway	Underpass	Side collision	Car	Young Adult	1	
165447	Idaho	########	Saturday	10:35:57	Snowy	Wet	Dusk	Freeway	Underpass	Head-on collision	Pedestrian	Young Adult	3	
165448	Florida	########	Friday	06:23:04	Rainy	Wet	Dawn	Urban	Underpass	Rollover	Car	Adult	4	
165449	Florida	########	Wednesday	16:26:10	Snowy	Dry	Night	Highway	Overpass	Rollover	Truck	Adult	3	
165450	Minnesota	########	Monday	02:11:27	Foggy	Snowy	Dusk	Highway	Overpass	Side collision	Motorcycle	Adult	3	
165451	Utah	########	Saturday	17:03:03	Foggy	Wet	Night	Highway	Intersection	Rollover	Bicycle	Senior	4	
165452	New Hamp	########	Monday	13:15:15	Sunny	Dry	Night	Highway	Crossroads	Run-off-road	Motorcycle	Young Adult	2	
165453	Kansas	########	Wednesday	05:13:23	Sunny	Icy	Daylight	Rural	Intersection	Head-on collision	Bicycle	Senior	1	
165454	Colorado	########	Monday	17:30:27	Snowy	Wet	Daylight	Street	Roundabout	Head-on collision	Car	Senior	1	
165455	Maryland	########	Thursday	16:48:46	Rainy	Wet	Dusk	Street	Roundabout	Head-on collision	Pedestrian	Teenager	3	
165456	Pennsylvar	########	Saturday	03:57:23	Rainy	Icy	Daylight	Rural	Crossroads	Side collision	Truck	Teenager	2	
165457	Oklahoma	########	Monday	12:36:33	Stormy	Icy	Dusk	Highway	Intersection	Rollover	Truck	Senior	3	
165458	Florida	########	Wednesday	06:53:50		Muddy	Dusk	Street	Underpass	Cyclist involved	Car	Adult	4	

Data Cleaning and Preparation

Before creating the dashboard, the dataset was cleaned and prepared to ensure accuracy and consistency in the analysis. The following data preprocessing steps were performed:

1. **Handling Missing Values**
   - Identified missing values in numerical value Speed Limit, Road Width, Vehicle Speed, Time Taken for Emergency Response.
   - Replaced or managed missing data with Average to avoid errors during analysis and visualization. Data Type Correction

2. **Data Type Correction**
   - Verified and corrected data types for relevant fields.
   - Converted columns such as Date into proper date format and ensured numerical fields were stored as numeric values.

3. **Rounding Decimal Values**
   - Rounded decimal values in numerical columns to improve readability and maintain consistency across reports and visualizations.

4. **Data Validation**
   - Reviewed the dataset after cleaning to ensure data accuracy and compatibility with Power BI visualizations.

Cleaned Dataset

The raw accident dataset was pre-processed to improve data quality before analysis. Missing values were handled, data types were corrected, and decimal values were rounded where necessary. These cleaning steps ensured data consistency, accuracy, and compatibility with Power BI visualizations, resulting in a reliable dataset for dashboard development and insight generation.

Screen short for raw uncleaned data is given below.

ID	State	Date	Day_of_Week	Time_of_Day	Weather_Conditions	Road_Conditions	Light_Conditions	Type_of_Road	Type_of_Junction	Type_of_Accident	Vehicle_
165433	Connecticut	20-02-2024	Tuesday	19:59:02	Rainy	Muddy	Daylight	Street	T-Junction	Run-off-road	Car
165434	Massachusetts	09-01-2024	Wednesday	21:28:51	Sunny	Muddy	Daylight	Highway	Intersection	Side collision	Pedestrian
165435	Wyoming	20-02-2024	Thursday	18:04:26	Snowy	Snowy	Dawn	Street	Underpass	Cyclist involved	Bicycle
165436	Hawaii	15-01-2024	Sunday	16:51:02	Snowy	Icy	Daylight	Street	Intersection	Run-off-road	Car
165437	Oklahoma	10-02-2024	Tuesday	14:19:19	Foggy	Icy	Night	Rural	T-Junction	Run-off-road	Car
165438	Minnesota	23-02-2024	Monday	00:14:21	Foggy	Snowy	Dawn	Urban	T-Junction	Head-on collision	Pedestrian
165439	Wyoming	19-02-2024	Saturday	10:21:34	Stormy	Icy	Daylight	Street	Crossroads	Rollover	Pedestrian
165440	Minnesota	25-01-2024	Tuesday	02:01:40	Rainy	Icy	Daylight	Urban	Underpass	Rear-end collision	Motorcycle
165441	Nevada	07-01-2024	Wednesday	06:09:08	Rainy	Wet	Dusk	Highway	Overpass	Rear-end collision	Car
165442	North Dakota	01-01-2024	Tuesday	22:33:51	Sunny	Icy	Dawn	Street	Overpass	Pedestrian involved	Pedestrian
165443	North Dakota	27-01-2024	Tuesday	05:49:45	Rainy	Wet	Dawn	Highway	Roundabout	Rear-end collision	Car
165444	Iowa	26-01-2024	Saturday	12:48:53	Stormy	Muddy	Night	Rural	Underpass	Rollover	Truck
165445	Rhode Island	22-02-2024	Tuesday	08:18:51	Sunny	Icy	Daylight	Freeway	T-Junction	Rollover	Pedestrian
165446	Delaware	05-01-2024	Monday	21:51:12	Sunny	Icy	Dawn	Highway	Underpass	Side collision	Car
165447	Idaho	19-02-2024	Saturday	10:35:57	Snowy	Wet	Dusk	Freeway	Underpass	Head-on collision	Pedestrian
165448	Florida	16-02-2024	Friday	06:23:04	Rainy	Wet	Dawn	Urban	Underpass	Rollover	Car
165449	Florida	11-01-2024	Wednesday	16:26:10	Snowy	Dry	Night	Highway	Overpass	Rollover	Truck
165450	Minnesota	20-01-2024	Monday	02:11:27	Foggy	Snowy	Dusk	Highway	Overpass	Side collision	Motorcycle
165451	Utah	09-01-2024	Saturday	17:03:03	Foggy	Wet	Night	Highway	Intersection	Rollover	Bicycle
165452	New Hampshire	05-02-2024	Monday	13:15:15	Sunny	Dry	Night	Highway	Crossroads	Run-off-road	Motorcycle
165453	Kansas	18-02-2024	Wednesday	05:13:23	Sunny	Icy	Daylight	Rural	Intersection	Head-on collision	Bicycle
165454	Colorado	25-02-2024	Monday	17:30:27	Snowy	Wet	Daylight	Street	Roundabout	Head-on collision	Car
165455	Maryland	22-02-2024	Thursday	16:48:46	Rainy	Wet	Dusk	Street	Roundabout	Head-on collision	Pedestrian
165456	Pennsylvania	27-01-2024	Saturday	03:57:23	Rainy	Icy	Daylight	Rural	Crossroads	Side collision	Truck
165457	Oklahoma	19-02-2024	Monday	12:36:33	Stormy	Icy	Dusk	Highway	Intersection	Rollover	Truck
165458	Florida	01-01-2024	Wednesday	06:53:50	Unknown	Muddy	Dusk	Street	Underpass	Cyclist involved	Car

DAX Measures & Calculations

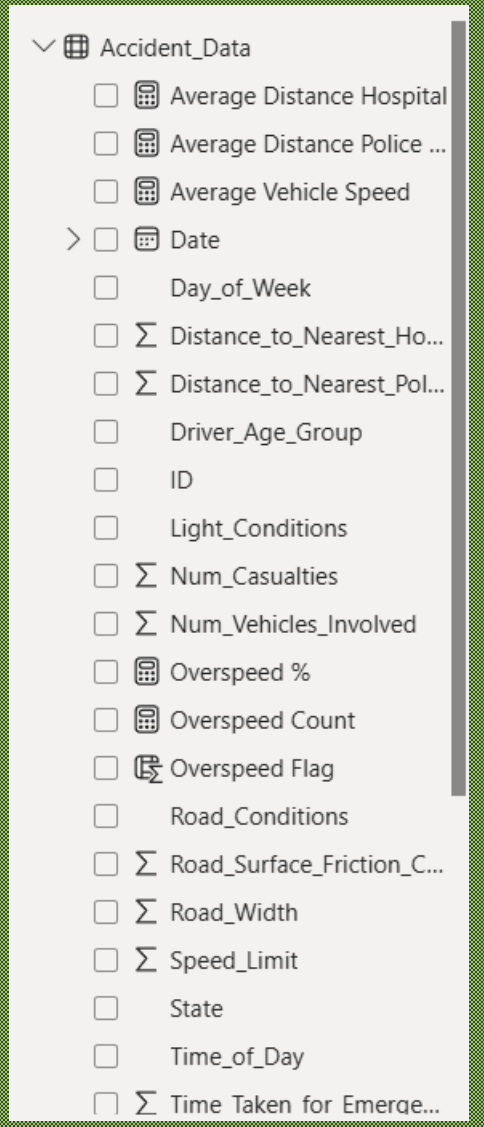
Data Analysis Expressions (DAX) were used in Power BI to create calculated measures that support dashboard analysis and KPI reporting. These measures enabled dynamic calculations based on user selections and filters.

The DAX process involved:

- Calculated total casualties across all accident records.
- Calculated total accident records.
- Calculated average vehicle speed.
- Generating summary metrics for dashboard KPI cards.
- Creating measures to calculate total accidents.
- Calculated Column Created:

Overspeed Flag: Identifies whether a vehicle was exceeding the speed limit at the time of the accident.

```
Overspeed Flag = IF( [Vehicle_Speed] > [Speed_Limit], "Overspeed", "Within Limit" )
```



Data Visualization Using Power BI

ROAD ACCIDENT ANALYSIS & SAFETY INSIGHTS

Comprehensive overview of road accidents to identify key patterns and improve safety

State: All

FILTERS

Day of Week: All

Light Conditions: All

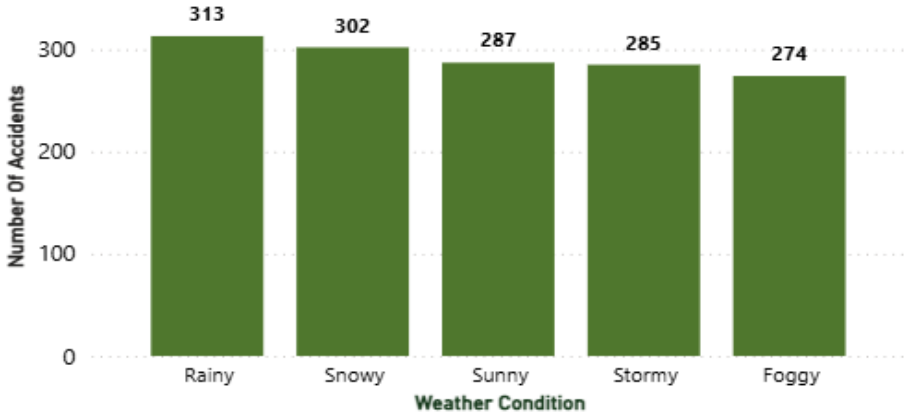
Type Of Road: All

Road Conditions: All

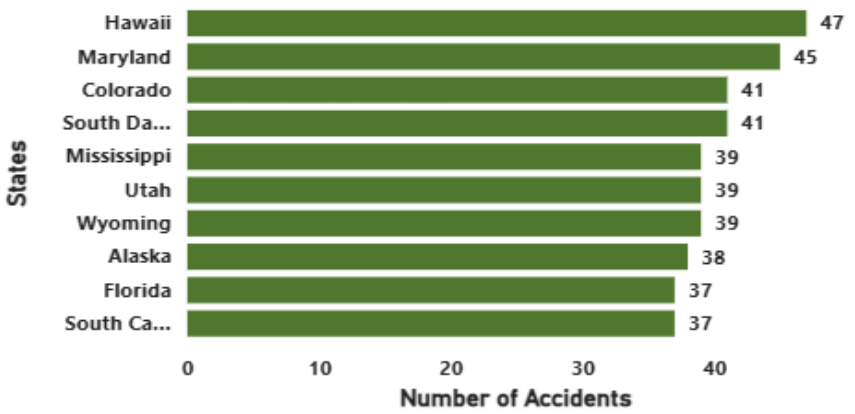
Diver Age Group: All

TOTAL ACCIDENTS	TOTAL CASUALITIES	AVG. VEHICLE SPEED	Overspeed %	AVG. DISTANCE TO HOSPITAL	AVG. DISTANCE TO POLICE STATION
1610	2325	49.94	48%	5.06	5.05

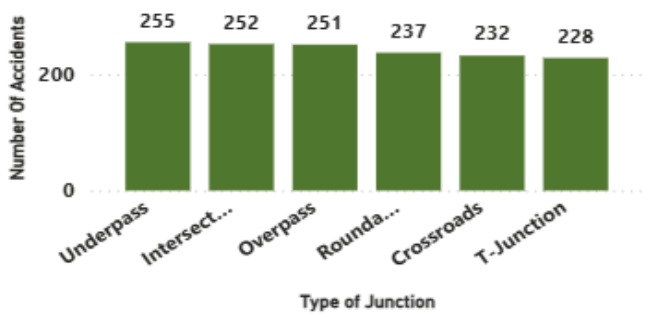
Accidents by Weather Conditions



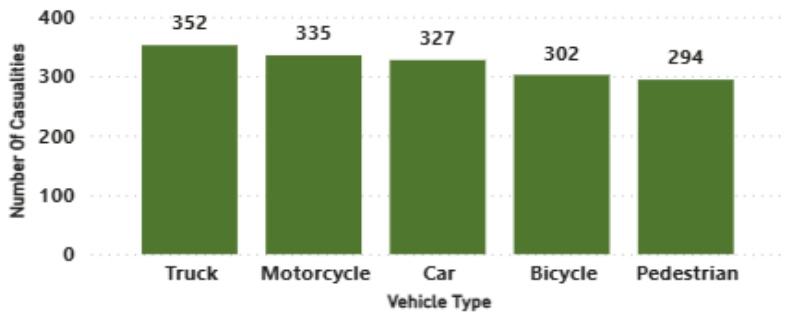
Accidents by State



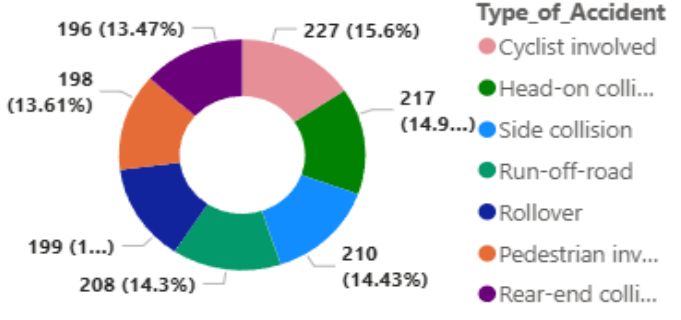
Accidents by Type of Junction



Casualties By Vehicle Type



Accidents by Type



Dashboard Analysis

Before creating the dashboard, the dataset was cleaned and prepared to ensure accuracy and consistency in the analysis. The following data preprocessing steps were performed:

1. Accidents by Weather Conditions

Rainy weather recorded the highest number of accidents, indicating that reduced road grip and visibility may increase accident risk.

2. Accidents by States

Hawaii , Maryland, Colorado, South Dakota

These states reported comparatively higher accident counts within the dataset.

3. Accidents by Type of Junction

Underpasses recorded the highest number of accidents, closely followed by intersections and overpasses.

4. Casualties by Vehicle Type

Truck-related accidents resulted in the highest casualty count. Due to their larger size and weight, truck accidents may lead to more severe consequences.

5. Accidents by Type of Accident

The distribution is relatively balanced, with each accident type contributing between approximately 13% and 16% of total accidents.

No single accident type dominates the dataset.

Overall Insights

1. Speeding contributes to a significant proportion of accidents, with 48% involving over speeding vehicles.

2. Rainy and snowy weather conditions show the highest accident frequencies.

3. Truck-related accidents produce the greatest number of casualties.

4. Certain states experience higher accident counts and may require targeted interventions.

5. Accidents occur across all junction types, indicating the need for comprehensive traffic management.

6. Accident types are evenly distributed, suggesting that multiple risk factors contribute to road accidents.

Thank You